Solution

$R_0 = 3\ \Omega, \quad T = 100^0C, \quad T_0 = 20^0C$

$\alpha = 0.004/^0C, \quad R_T = ?$

$R_T = R_0(1 + \alpha(T\text{-}T_0))$

$R_{100} = 3(1 + 0.004 \times 80)$

$R_{100} = 3.96\ \Omega$

EXAMPLE 2.14

Resistance of a material at 20^0C and 40^0C are 45 Ω and 85 Ω respectively. Find its temperature coefficient of resistivity.

Solution

$T_0 = 20\ ^0C, T = 40\ ^0C, R_0 = 45\ \Omega, R = 85\ \Omega$

$$\alpha = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T}$$

$$\alpha = \frac{1}{45}\left(\frac{85-45}{40-20}\right) = \frac{1}{45}(2)$$

$$\alpha = 0.044\ \text{per}^\circ C$$

2.3 ENERGY AND POWER IN ELECTRICAL CIRCUITS

When a battery is connected between the ends of a conductor, a current is established. The battery is supplying energy to the device which is connected in the circuit. Consider a circuit in which a battery of voltage V is connected to the resistor as shown in Figure 2.15.

Assume that a positive charge of dQ moves from point a to b through the battery and moves from point c to d through the resistor and back to point a. When the charge

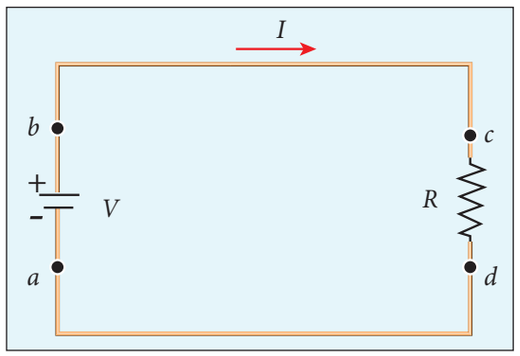


Figure 2.15 Energy given by the battery

moves from point a to b, it gains potential energy $dU = V.dQ$ and the chemical potential energy of the battery decreases by the same amount. When this charge dQ passes through resistor it loses the potential energy $dU = V.dQ$ due to collision with atoms in the resistor and again reaches the point a. This process occurs continuously till the battery is connected in the circuit. The rate at which the charge loses its electrical potential energy in the resistor can be calculated.

The electrical power P is the rate at which the electrical potential energy is delivered,

$$P = \frac{dU}{dt} = \frac{(V.dQ)}{dt} = V\frac{dQ}{dt} \qquad (2.31)$$

Since the electric current $I = \frac{dQ}{dt}$, the equation (2.31) can be rewritten as

$$P = VI \qquad (2.32)$$

This expression gives the power delivered by the battery to any electrical system, where I is the current passing through it and V is the potential difference across it. The SI unit of electrical power is watt ($1W = 1\ J\,s^{-1}$). Commercially, the electrical bulbs used in houses come with the power and voltage rating of 5W-220V, 30W-220V, 60W-220V etc. (Figure 2.16).

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